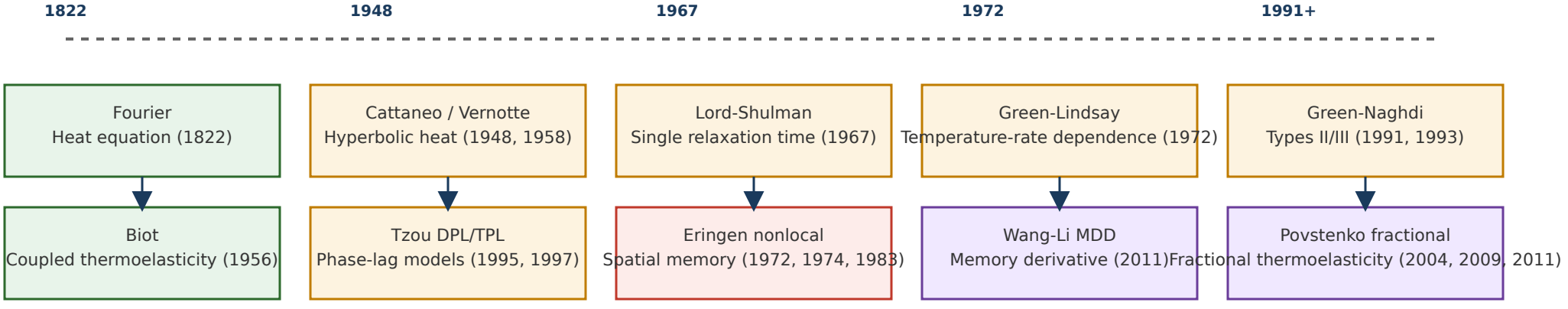
Theory family evolution in generalized continuum mechanics

Historical development of major theory families from classical to modern multiphysics formulations (Section 2)



Classical	Generalized	Nonlocal	Fractional
Fourier heat conduction Classical coupled thermoelasticity (Biot, 1956) Infinite thermal speed	LS, GL, GN-II/III DPL, TPL, MGT Model-dependent propagation Relaxation times	Eringen nonlocal Strain-gradient Size-dependent waves Spatial memory	MDD, fractional Thermoviscoelastic Hybrid models Temporal memory

Reading the figure

Vertical arrows group selected milestones; they do not assert direct derivation or equivalence.
Family groupings and cross-family relationships remain provisional and require source checking.
No cross-family reduction or convergence relation is asserted by this schematic.
Bottom boxes group theories by their dominant physical mechanism.
Propagation and support properties depend on the exact constitutive equations.
Source: inherited literature inventory; dates and relationships are not fully source-checked.